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SESSIONAL PAPER

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SECTION A ($5 \times 1 = 5$ marks)

- A. Discuss the advantages of Multiprogramming.
- B. Define concurrency principle.
- C. Define semaphore and its types.
- D. Explain asymptotic notations used in algorithm analysis.
- E. State and explain the two primary functions of an Operating System.

SECTION B (Attempt any one from each)

- Q2. (A) Explain circular queue and insertion code.
(B) Explain producer consumer problem.
- Q3. (A) Discuss Multiuser Systems and Multithreaded Systems in detail.
(B) Explain mutual exclusion and critical section problem.
- Q4. (A) Differentiate between Time-Sharing and Multiprogramming Operating Systems.
(B) Differentiate between Monolithic Kernel and Microkernel Architecture.
- Q5. (A) Explain polynomial representation using linked list.
(B) Calculate address of $A[10][20][30]$ in 3D array.
- Q6. (A) Define Operating System and explain its major functions.
(B) Explain Multiprocessor OS, Real-Time OS, and Time-Sharing OS.